

BENG 285/BNFO 285/ECE 204. Statistical Learning in Bioinformatics

Project 3: Unsupervised machine learning for deciphering the structure of multidimensional datasets.

Learning objectives: The overall learning objective of this project is to utilize a nonnegative dimensionality reduction approach to decompose a large multi-dimensional dataset into a set of meaningful and understandable features. Specifically, the project will require students to examine the provided dataset as a blind source separation problem and to solve this problem using nonnegative matrix factorization (NMF) or an approach analogous to NMF. Students will need to consider selecting the appropriate number of nonnegative components based on their overall stability and their ability to accurately describe the provided dataset. Students will also likely need to utilize nonparametric and/or parametric statistical tests to evaluate whether any of the derived nonnegative factors are enriched and/or depleted within samples based on the available patients' metadata.

Subject Matter: The focus in project 3 will be the detection of mutational signatures in cancer genomes. The project will require students to construct a theoretical model that explains the set of somatic mutations within a cancer genome as a mixture of distinct nonnegative additive processes. Students will need to apply nonnegative matrix factorization (NMF) to their provided datasets by first converting the somatic mutations to a mutational matrix. An important part of the project is determining the most meaningful set of mutational signatures that described the cancer mutational data. Additionally, students will need to determine whether any of the identified nonnegative factors are biologically and/or clinically meaningful by evaluating for any statistically significant differences in regards to patients' metadata (*e.g.*, cancer subtype, age of diagnosis, sex, race, clinical endpoint, center generating the data, gene expression data, *etc.*). Lastly, students will need to compare the derived nonnegative factors to the ones generated by applying other recently published approaches.

Project's manuscript: [https://www.cell.com/cell-genomics/fulltext/S2666-979X\(22\)00124-0](https://www.cell.com/cell-genomics/fulltext/S2666-979X(22)00124-0)

Manuscript presentation: Team presentation on Tuesday (6-May) at 6:30pm (CENTR 217B)

Project's datasets: <https://www.dropbox.com/scl/fo/7d37xqur5v1b8jni61b0t/AC1lf0-cbGkGBSZFI9fXzdQ?rlkey=pfw7xmb7slnz7d398gzfzpu7>

Project's presentation: Team presentation on Thursday (15-May) at 6:30pm (CENTR 217B)

Deadline: All teams submit a project report (max 3 pages) by 11:59pm on Friday (16-May)

Potential outline for approaching the project:

- Convert the set of somatic mutations into a mutational matrix based on previously established classifications.
- Apply an existing NMF algorithm for decomposing the data into a specific number of nonnegative components.
- Evaluate the appropriate parameters for running NMF including initialization, matrix normalization, and selection of an objective function.
- Determine the optimal number of nonnegative components present in your dataset based on their overall stability and their ability to accurately describe the provided dataset.
- Compare your results to the ones from directly applying one or more of these recently published approaches: SigProfilerExtractor, SignatureToolsLib, MutationalPatterns, and/or Signer.